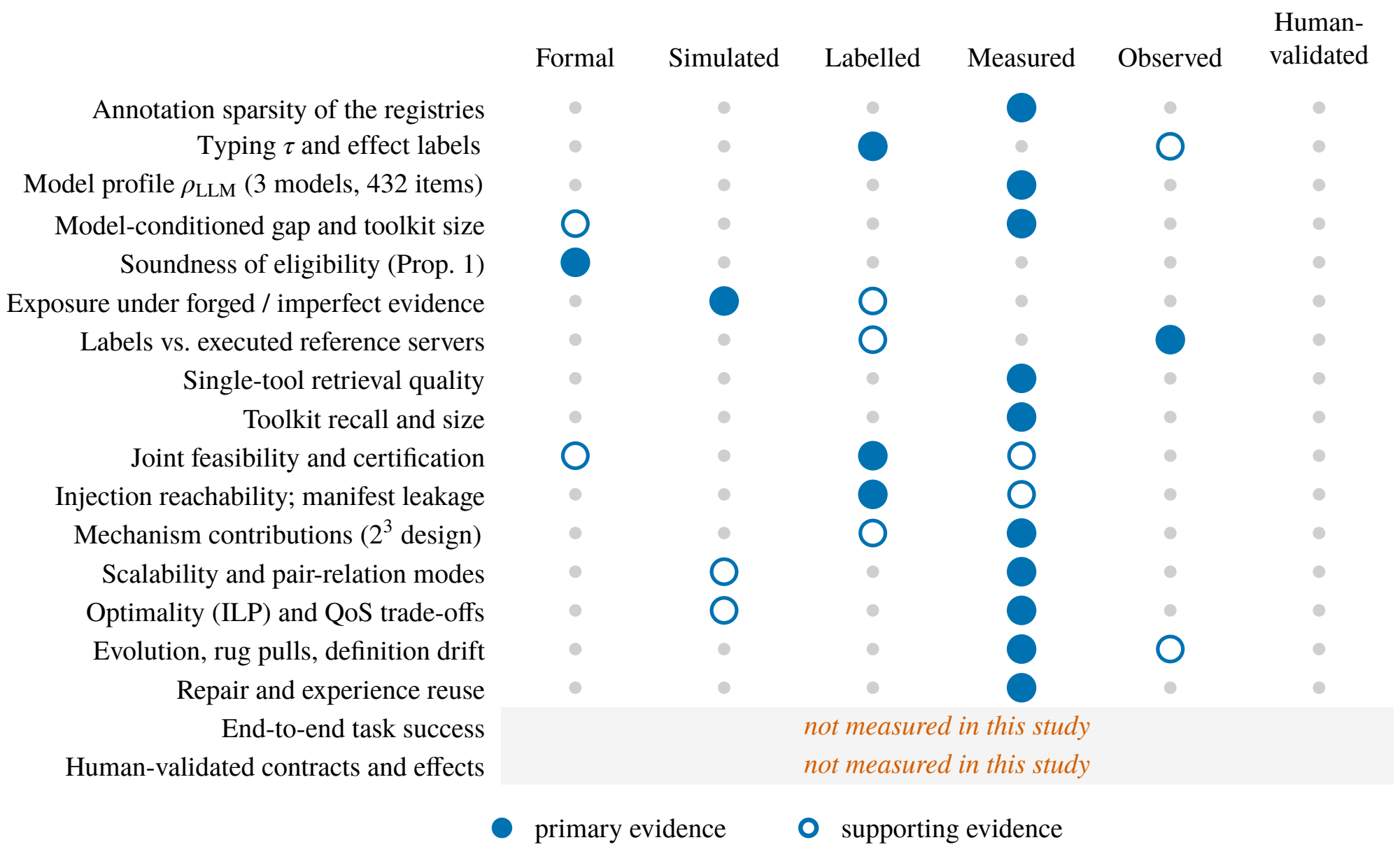
Formal
Simulated
Labelled
Measured
Observed
Human-validated
Annotation sparsity of the registries
Typing τ and effect labels
Model profile ρ_{LLM} (3 models, 432 items)
Model-conditioned gap and toolkit size
Soundness of eligibility (Prop. 1)
Exposure under forged / imperfect evidence
Labels vs. executed reference servers
Single-tool retrieval quality
Toolkit recall and size
Joint feasibility and certification
Injection reachability; manifest leakage
Mechanism contributions (2^3 design)
Scalability and pair-relation modes
Optimality (ILP) and QoS trade-offs
Evolution, rug pulls, definition drift
Repair and experience reuse
End-to-end task success
not measured in this study
Human-validated contracts and effects
not measured in this study
primary evidence
supporting evidence